



NORTHWIND TRADERS

Northwind Order Management System

System Architecture · v2.0 · 2026

● Production Ready

● Clean Architecture · DDD

● Containerized · 3-tier

● Internship · 2026

01 CLIENT

Browser · SPA

localhost:9000 / orders

V Vue 3 SPA · Composition API

- orders Inventory-aware order forms
- analyt KPI cards · bar & donut charts
- pdf Invoice preview & download
- xlsx Excel export

State & data

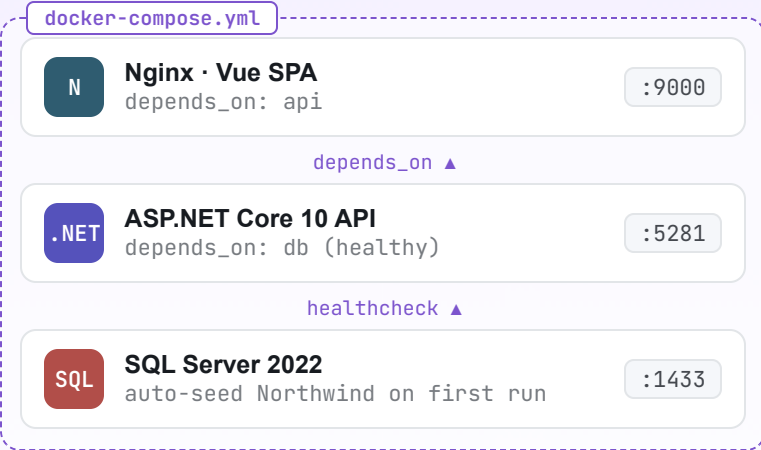
pinia · axios

- Pinia stores per aggregate (orders, products, customers)
- Axios client with ProblemDetails error mapping

vue3 quasar pinia axios apexcharts
sheetjs

02 DOCKER INFRASTRUCTURE

Compose · 3 containers

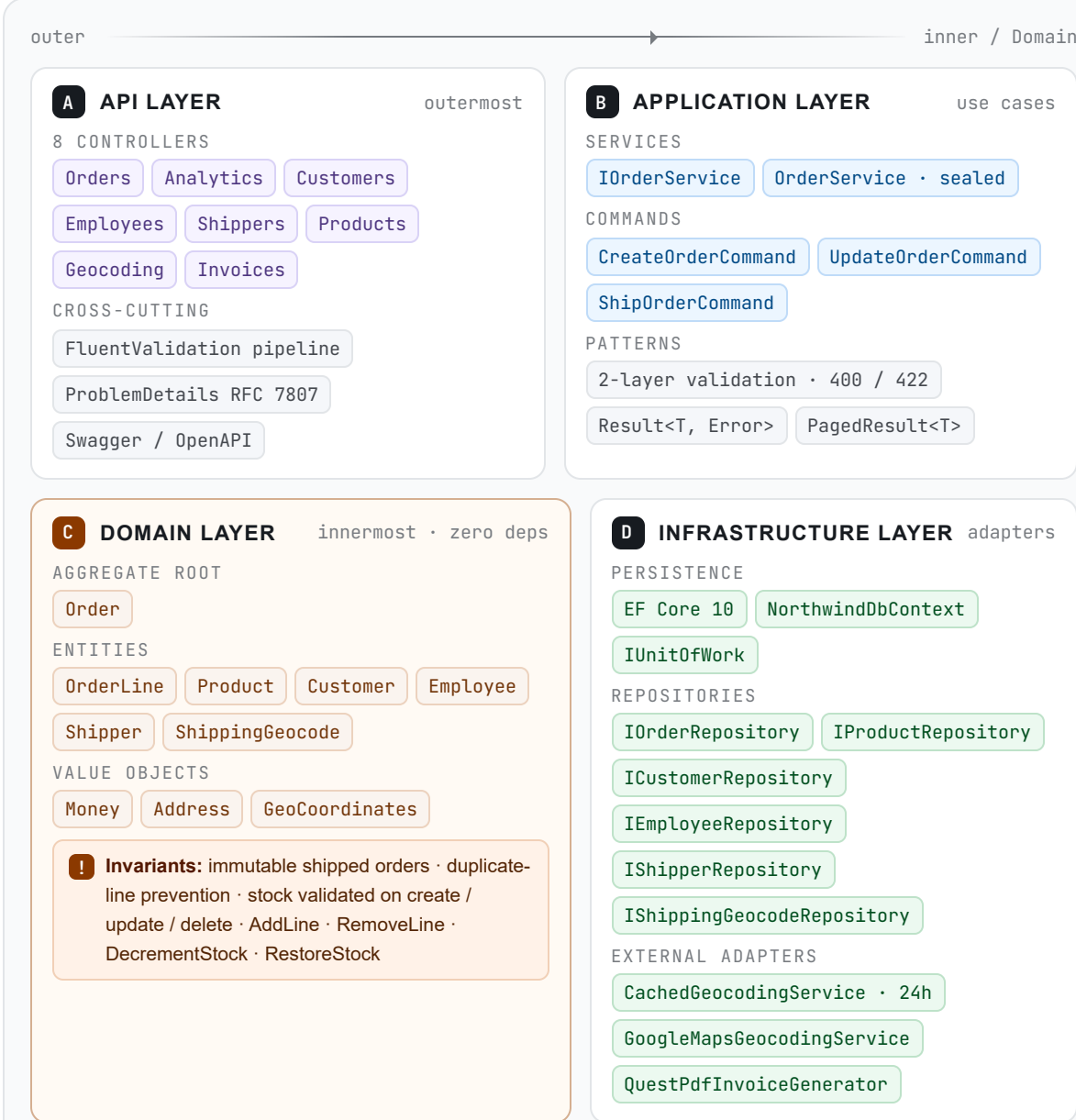


- CORS
- Swagger / OpenAPI
- ProblemDetails RFC 7807
- Health checks

docker-compose nginx sql-server-2022 .NET 10
cors swagger

03 CLEAN ARCHITECTURE

4 layers · dependencies point inward → Domain



04 DATA & EXTERNAL

Persistence · 3rd-party APIs

- SQL SQL Server 2022 · Northwind
- Legacy schema preserved · 0 changes
 - Custom ShippingGeocodes table via additive migration

- G Google Maps Platform
- Geocoding API — address → coordinates
 - Maps JavaScript API — interactive map UI
 - Maps Static API — invoice thumbnails

- PDF QuestPDF
- A4 branded invoices · line items · subtotals
 - Embedded delivery map thumbnail

QUALITY & TEST COVERAGE

xUnit · Coverlet · ReportGenerator

● 202

tests passing

90.2%

line coverage

80.3%

branch coverage



Measured: Domain · Application · Api — Infrastructure excluded (EF Core / HTTP not unit-testable).

KEY DESIGN DECISIONS

Result<T, Error> over exceptions
Explicit failure paths · no surprise throws in business logic

Order as aggregate root
Invariants enforced through the aggregate · no entity bypass

ShippingGeocodes separate table
Legacy Northwind schema untouched · additive migration only

IsShipped derived from ShippedDate
Single source of truth · no extra status column to drift

Trade-offs & principles enforced across the codebase

CachedGeocodingService decorator
Open / Closed principle · 24h cache transparent to callers

IOrderService interface
Controllers decoupled from implementation · fully mockable